🧠 🔄 COMPLETE REAL FLOW (UNDERSTAND FIRST)

Create Barcode → Insert Same ID in DB → Scan → Match → Action

👉 VERY IMPORTANT RULE:

> **Barcode value MUST match database ID exactly**

📌 1. HOW TO GENERATE BARCODE (STEP-BY-STEP)

123456789012

Interleaved 2 of 5 Industrial

123456789012

Interleaved 2 of 5 Matrix

123456789012

Interleaved 2 of 5

123456789012

CODE-128

123456789012

EAN-128

⚠️

⚠️

🔹 OPTION 1: Online (EASY)

Use:
👉 Barcode Generator

Steps:

1. Open website
2. Select:
    - Type → **Code128** (best for your project)
3. Enter value:

B001

4. Click **Generate**
5. Download image
6. Print it

🔹 For USERS

U001
U002

🔴 IMPORTANT RULE

Thing	Example

Book Barcode	B001
User Barcode	U001

👉 SAME value must be in database

🗄️ 2. CREATE DATABASE (FROM ZERO AGAIN)

🔹 Step 1: Open MySQL

sudo mysql -u root -p

🔹 Step 2: Create DB

```
CREATE DATABASE library_db;
USE library_db;
```

🔹 Step 3: Create Tables (FINAL CORRECT)

```
CREATE TABLE users (
   id VARCHAR(50) PRIMARY KEY,
   name VARCHAR(100)
);

CREATE TABLE books (
   id VARCHAR(50) PRIMARY KEY,
   name VARCHAR(100),
   available BOOLEAN DEFAULT TRUE
);

CREATE TABLE transactions (
   id INT AUTO_INCREMENT PRIMARY KEY,
   user_id VARCHAR(50),
```

```
    book_id VARCHAR(50),
    issue_date DATETIME,
    return_date DATETIME,
    fine FLOAT DEFAULT 0
);
```

📥 3. INSERT DATA (VERY CLEAR)

🔹 Insert Users

```
INSERT INTO users VALUES ('U001','Atharv');
INSERT INTO users VALUES ('U002','Rahul');
```

🔹 Insert Books

```
INSERT INTO books VALUES ('B001','Python Book',1);
INSERT INTO books VALUES ('B002','DSA Book',1);
```

🔍 VERIFY

```
SELECT * FROM users;
SELECT * FROM books;
```

🔗 4. HOW DATABASE CONNECTS TO MAIN CODE

🔹 Create file: `database.py`

```
import mysql.connector

def connect_db():
    return mysql.connector.connect(
        host="localhost",
        user="root",
        password="YOUR_PASSWORD",
        database="library_db"
    )
```

📷 5. HOW BARCODE CONNECTS TO DATABASE

🔹 Scanner returns:

```
barcode_data = "B001"
```

🔹 Then main code checks:

```
cursor.execute("SELECT * FROM books WHERE id=%s", (barcode_data,))
```

🔴 IMPORTANT LOGIC

Barcode	Query	Result
B001	SELECT book	Found
U001	SELECT user	Found

📁 6. FINAL FILE STRUCTURE (VERY IMPORTANT)

```
library_system/
│
├── main.py          ⭐ MAIN FILE
├── database.py      ⭐ DB connection
├── scanner.py       Barcode scan
├── lcd.py           LCD display
├── sensors.py       IR sensors
├── admin_panel.py   GUI
```

🔗 7. HOW FILES CONNECT (IMPORTANT)

🔹 Inside `main.py`

```
from database import connect_db
from scanner import scan_barcode
from lcd import lcd_display
from sensors import detect_mode
```

👉 This is how modules connect

🧠 8. MAIN EXECUTION FLOW (REAL)

🔹 Step-by-step

1. Run:

```
python3 main.py
```

---

2. LCD shows:

Select Mode

---

3. IR Sensor Trigger

👉 ISSUE / RETURN selected

---

4. Scan Book

👉 Barcode → "B001"

---

5. System checks DB

SELECT * FROM books WHERE id='B001';

---

6. Scan User

👉 "U001"

---

7. System processes:
- Issue → update DB
- Return → calculate fine

⚠️ 9. MOST IMPORTANT THINGS TO REMEMBER

🔴 1. Barcode = Database ID

👉 MUST match EXACTLY

🔴 2. File Names Matter

File	Purpose
main.py	Main system
database.py	DB connect
scanner.py	Scan

🔴 3. Database Name MUST match

database="library_db"

🔴 4. Table Names MUST match

users
books
transactions

🔴 5. Column Names MUST match

id, name, available

🔴 6. Password MUST be correct

🧪 10. FULL TEST (REAL SCENARIO)

🔹 You scan:

Book → B001
User → U001

🔹 System does:

SELECT * FROM books WHERE id='B001';

✔ Found

UPDATE books SET available=0

🔹 Output:

Issued Successfully